Practice Quiz: Agents — Neural Correlates of the Self-Model

Name: __ ***Date:*** __

Part A: Multiple Choice

1. The default mode network (DMN) is proposed to implement: A) Motor control during complex tasks B) The neural substrate of the self-model — the brain's most abstract, temporally extended predictions about oneself C) Primary sensory processing during passive observation D) Cerebellar coordination of movement

2. An emotion, according to interoceptive inference, is: A) A pre-programmed response stored in the limbic system B) An interoceptive prediction error — the discrepancy between predicted and actual bodily states C) A purely cognitive judgment with no bodily component D) Random neural noise that the brain interprets post-hoc

3. The rubber hand illusion demonstrates: A) That humans are easily tricked by visual stimuli B) That body ownership is an inference — a posterior probability updated by multisensory evidence — not a fixed property C) That the brain ignores proprioceptive information D) That visual information always overrides touch

4. In depersonalization, the Active Inference mechanism is: A) Excessive interoceptive precision causing overwhelming emotions B) Low precision on interoceptive prediction errors — the "aliveness signal" is attenuated, producing feelings of unreality C) Hyperactive DMN producing excessive self-referential thought D) Normal interoceptive processing with delusional overlay

5. Cotard's syndrome (believing one is dead) is best explained by: A) Brain death affecting cognition B) Catastrophic interoceptive inference failure — the self-model receives so little interoceptive evidence that the best Bayesian hypothesis is "I am dead" C) Frontal lobe damage causing confabulation D) Malingering for attention

6. The anterior insula is critical for: A) Language production (Broca's area) B) Integrating interoceptive signals with emotional, cognitive, and social context — implementing the "sentient self" C) Visual processing of faces D) Spatial navigation

7. Alexithymia (difficulty identifying emotions) is associated with: A) Damage to the motor cortex B) Reduced interoceptive precision — the person cannot reliably distinguish interoceptive prediction errors, so emotions feel undifferentiated C) Enhanced cognitive processing at the expense of emotional processing D) Normal interoceptive processing with a language deficit

8. During anesthesia, which self-model component is disrupted first? A) Motor planning — patients stop moving B) Interoceptive and DMN self-referential processing — the self-model is lost before specific sensory content processing C) Visual processing — patients go blind D) Memory consolidation — patients can't form new memories

Part B: Short Answer

1. A patient with somatoparaphrenia (following right parietal stroke) looks at her own left arm and says: "That's not my arm. Someone put a stranger's arm in my bed." Explain this using Active Inference: What component of the self-model has failed? Why does the brain prefer "this is not my arm" over "my brain has a deficit"? (Hint: the prior that "I am intact" has very high precision.) (250 words)

2. Experienced meditators show reduced DMN activity and report dissolution of the sense of a fixed self. Does this mean meditation is "brain damage"? Argue why reducing self-model precision is fundamentally different from pathological self-model disruption (depersonalization). What is the key difference? (250 words)

Part C: Essay

1. The "minimal self" (pre-reflective bodily awareness) and the "narrative self" (autobiographical identity) dissociate in several clinical conditions. Amnestic patients (like H.M.) lose the narrative self but retain the minimal self. Depersonalization patients retain the narrative self but lose the minimal self. Write a 400-word analysis explaining each case in

Active Inference terms, mapping each to specific neural substrates and precision parameters. What does this double dissociation reveal about the architecture of the self?

2. A colleague claims: "The self is obviously real — you can't deny your own existence. Active Inference's claim that the self is 'just a prediction' is absurd." Write a 400-word response that: (a) acknowledges why the self *feels* real and certain, (b) explains why Active Inference's claim is not that the self is "unreal" but that it is an actively maintained inference, and (c) provides three clinical examples where the self-model systematically fails, showing that selfhood IS indeed an inference that can go wrong.